

Science

Advanced Nature Study Lessons
Advanced Nature Study Afternoon Work





About the Course

This course includes the following topic(s): Advanced Nature Study Lessons, Advanced Nature Study Afternoon Work

About Advanced Nature Study Lessons

An elective in Environmental Science, Advanced Nature Study requires independent interest as it is designed to be personalized by the student and requires some flexibility in scheduling due to the amount of fieldwork required beyond the regular lesson time.



Placement & Combining Tips

Advanced Nature Study Lessons

Students should have at least 3 years of nature study experience and be comfortable with their nature journal.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Advanced Nature Study Lessons 5 times/week 45 min	Master of Field Arts
9-12	Advanced Nature Study Afternoon Work 1+ time/week 20 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Advanced Nature Study				
Advanced Nature Study Lessons	Advanced Nature Study Lessons	Advanced Nature Study Lessons	Advanced Nature Study Lessons	Advanced Nature Study Lessons Advanced Nature Study Afternoon Work



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Advanced Nature Study Lessons

Download, save, bookmark, or print a copy of How to Teach Nature Journaling for regular use.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Science: Advanced Nature Study

Advanced Nature Study Lessons



Master of Field Arts



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Science: Advanced Nature Study



Clear Perspex Palette with Magnet Strip and Dry-Erase Marker



Quick Links

Science: Advanced Nature Study

Advanced Nature Study Lessons

- ∞ [Seek app from iNaturalist](#)
- ∞ [SHOP — Exploring Overland](#)
- ∞ [How to Teach Nature Journaling](#)
- ∞ [A Community for Naturalists · iNaturalist](#)

[Click THIS text or scan the QR code for links.](#)



Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1

45m Advanced Nature Study Lessons - Lesson 1 Sections 1-3

Materials: Master of Field Arts, Journal

→ INTRO

As we start our nature journaling adventure, it is important to remember local, state, and federal laws about collecting and specific concerns when in the field. Look up any local or state laws that you need to be aware of.

→ READ

- ∞ Article Link: Leave No Trace (7 Principles)
- ∞ Article Link: Can I Keep This?

→ INTRO

As we begin working through our Master of Field Arts book, we will start with a look at various tools that Roseann recommends. You may already have some of these tools or a variation of them. You may already use some or all of these tools in your nature study. The goal is to gather new ideas for tools and practices. There is only one tool that you were required to purchase for this course, and that was the Perspex Palette (see the supply list), as we will be using it quite a bit. For the rest of the items, you are free to evaluate and decide what you want to add to aid you in further observation and study. You can look up items mentioned in the chapter, and there will be links to additional or similar items, at times included in the lesson.

→ READ

p.7-17

→ LOCATE

Check to see if you already own any of the items mentioned or something similar. If not, consider whether you want to purchase any items for further exploration. You may want to try out your phone's built-in magnifier. Search in your Apps to find the magnifier and look up how to use it.

→ VIEW

Take a look at some other interesting historical field notebooks.
∞ Article Link: Field Notebooks - Tumamoc Sketchbook

→ JOURNAL

If you have any time left, use some form of magnification like a loupe, magnifying glass, binoculars, microscope, or your phone to view something from nature and replicate it in your journal.

• EXTRA HELPINGS

If interested, take a look at a loupe and these binoculars.
∞ **Item Link:** 10x and/or 30x (good for plant parts) Loupe (This item is from Amazon, but also take a look at Roseann's Discovery Kit on her website, which includes a basic loupe. Don't forget about her generous discount code offering 25% off any products in her store, which you can use throughout this course.)
∞ **Item Link:** Close Focus Binoculars recommended by John Muir Laws

WEEK 1

45m Advanced Nature Study Lessons - Lesson 2 Sections 1-3

Materials: How to Teach Nature Journaling, Journal

→ INTRO

Today is a journaling day. A prompt from John Muir Laws' book, How to Teach Nature Journaling, will provide some focus for today's journaling. You will be introduced to one or two prompts each week to give you many observation tools to use in your nature study. Use the link in the quick links to download a free digital copy of the book or purchase a paperback copy if you prefer.

As you read the instructions each time, note the helps in teaching these skills to others. Be thinking about someone you could introduce nature journaling to, and/or share the joys of nature journaling with, and look for opportunities to show them what you are learning. It might be a family member, a neighbor, or a friend.

→ READ OR VIEW

- p.36-38 (in How to Teach Nature Journaling)
- ∞ Video Link: I Notice, I Wonder...

→ JOURNAL

Head outside or bring something inside to practice INIWIRMO on (I notice, I wonder, It reminds me of).

• EXTRA HELPINGS

If interested, take a look at Roseann's homemade leather nature journal cover. Maybe you want to make one for yourself!
∞ **Article Link:** Roseann's nature journal cover

WEEK 1

45m Advanced Nature Study Lessons - Lesson 3 Sections 1-3

Materials: Master of Field Arts

→ READ

p.18-26 skip p.25 as we will come back to this later. We will spend some dedicated time later in the course on collecting and drawing tracks in your journal

→ LOCATE

There are many measuring and collecting options mentioned in the reading. See what you can find at home to use. Then, think through what you might want to purchase to start using in the field.
A pocket knife may be in a kitchen drawer, or a family member may have one that you could use.

• EXTRA HELPINGS

If interested, here are a few links to a couple of the items mentioned in the reading.
∞ **Link:** Pocket Net
∞ **Link:** Kestrel weather stations
∞ **Link:** Morakniv Clipper
∞ **Link:** Nature Journaling Field Bag (Could be a great Christmas or birthday gift idea!)
∞ **Article Link:** Preserving skulls and insects

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

Also, consider what options are on your phone for measuring heights and widths as well as angles. Find the Measure App and look up how to use it. Practice using it for heights as well as angles.

The End Notes start on p.159 and have many links for the items mentioned in the reading. You will need to figure out how you want to carry these items while you are in the field. See what bag options you have available and start taking your tools with you!

WEEK 1 45m Advanced Nature Study Lessons - Lesson 4

Materials: Journal

→ JOURNAL

Head outside and use the tools that you have collected to start investigating, observing, and journaling!

• EXTRA HELPINGS

If interested, here is a deep dive into choosing and using binoculars and scopes.

∞ **Video Link:** Binoculars

WEEK 1 45m Advanced Nature Study Lessons - Lesson 5

Materials: Journal

→ JOURNAL

Enjoy!

WEEK 1 20m Advanced Nature Study Afternoon Work - Lesson 1

→ FIELD WORK

Don't forget to schedule an extended time of exploring and nature journaling sometime each week for the lab component of this course. The more pencil miles you put in, as John Muir Laws says, the better your sketching and journaling skills will get!

WEEK 2 45m Advanced Nature Study Lessons - Lesson 6

Sections 1-3

Materials: Master of Field Arts

→ INTRO

Let's step back for a moment to think about some term and school-year-long projects for this course. One of the joys of regular journaling is being able to see and observe seasonal changes. This is the study of phenology. Charlotte Mason had students create a Book of Firsts or add this component to their nature journals. Whenever you notice something for the first time in a season, such as the first frost, snowflakes, hurricane, plant blooms, bird, or butterfly, record the date on a perpetual calendar or in your nature journal. As you do this year after year, you start to notice patterns and deviations from patterns that are fascinating!

→ READ

p.157-158

→ PLAN

Take some time to think about what locations around you would make an interesting Phenology Study. Choose one or two places to visit each week for the whole school year. This might be in your yard, down the street, or somewhere not far from where you live. You want it to be accessible so that you will be sure to go there every week to record changes and capture what you are seeing. Record the changes observed via words, pictures, sketches, measurements, etc., in your journal. Follow the suggestions in your reading on p.158.

Then, choose a couple of items to do a term or single-season study, such as a single tree, plant, bird, or even a seed you grow in a pot. Take the time of year into account when choosing. This is described in your reading as a small or medium-scale study. But you are not limited to the examples in the book. Find something close to where you live that you are interested in studying. Choose when in the year you will do your studies. When you are ready, write down your plan for what you are studying and when in your week you will do your specific observations and journaling for your special studies. You might want to use a separate journal for your year-long study and term studies. It will be so interesting to read back over your journal entries for this one area or item and see the changes!

→ READ

Another aspect of advanced nature study that we want to be a part of now and perhaps into the future is some sort of citizen science project. Take some time to read about some potential studies that pertain to your area using the links provided. Find one that you want to participate in. Nature's Notebook might have a project that relates to what you chose for your phenological study. Sign up for the project that interests you and follow the directions to get set up to start. Your observations are important and valuable contributions to science!

∞ Article Link: 9 Citizen Science Projects

∞ Article Link: Citizen Science

∞ Article Link: 20 Citizen Science Projects

∞ Article Link: Nature's Notebook

• EXTRA HELPINGS

If you are interested in adding a Book of Firsts to your different journals, take a look at Nathaniel Wheelwright and Bernd Heinrich's beautiful notebook.

∞ **Link:** The Naturalist's Notebook

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 2	45m Advanced Nature Study Lessons - Lesson 7 Materials: Journal → JOURNAL Enjoy! If nothing on the citizen science sites seemed interesting, try Project Squirrel! ∞ Link: Project Squirrel	
WEEK 2	45m Advanced Nature Study Lessons - Lesson 8 <i>Sections 1-3</i> Materials: Master of Field Arts → INTRO Roseann is a seasoned naturalist and describes her method of taking field notes, journaling, and organizing her notes for later scientific study and writing projects. Don't get bogged down in the details of Part One (p.27-33). Reading about her process is helpful to see how a naturalist documents all of their work. Part 2, starting on p.34, will have some more practical information about metadata that you may want to incorporate into your journal pages. → READ p.27-40 There is no right way to keep a journal. The main idea is to work in your journal at least four times a week to grow in all the skills and wonder of seeing what God has made! Try out new ideas and see what works best for you.	
WEEK 2	45m Advanced Nature Study Lessons - Lesson 9 Materials: How to Teach Nature Journaling, Journal → INTRO Today is a journaling day. A prompt from John Muir Laws' book, How to Teach Nature Journaling, will provide some focus for today's journaling. → READ p.88, 89, 91 → JOURNAL Use the 5 Ws and 1 H to observe and journal today.	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 2	45m Advanced Nature Study Lessons - Lesson 10 Materials: Journal → VIEW ∞ Video Link: NJC Metadata → JOURNAL If you aren't already in the practice of adding metadata to your page, start today to add the date, time, location, weather, and any other identifiers you want to add.	
WEEK 2	20m Advanced Nature Study Afternoon Work - Lesson 2 → FIELD WORK Spend an extended time on field work.	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 11 <i>Section 1-3</i> Materials: Master of Field Arts, Journal → INTRO Today, we will start with the humble pencil! → READ p.41-46 → VIEW Watch John Muir Laws explain other ways to use relative measurements to get the proportions right on your page. ∞ Video Link: NJC Measured Drawings → JOURNAL Practice mapping out some objects with dots and using relative measurements to aid you in your sketches.	• EXTRA HELPINGS If interested, watch how a pencil is made! ∞ Video Link: Pencils

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 3	45m Advanced Nature Study Lessons - Lesson 12 Materials: Journal → JOURNAL Enjoy!	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 13 Materials: Paper, Pencil → VIEW Drawing is useful for more things than nature study. Watch this talk on how drawing helps you think. ∞ Video Link: How Drawing Helps You Think → READ & PRACTICE As you read through the article, practice each type of exercise so you will be able to incorporate these into your journaling routine. ∞ Article Link: 12 Simple Warm-Up Exercises	• PLAN WEEKLY special study observations citizen science observations extended journaling time
WEEK 3	45m Advanced Nature Study Lessons - Lesson 14 Materials: How to Teach Nature Journaling, Journal → READ p.64-66 → VIEW ∞ Picture Link: Durer's The Great Piece of Turf from 1503, measuring 40.8 x 31.5 cms → JOURNAL Set up your sting and start recording what you find!	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 15 Materials: Journal → JOURNAL Enjoy!	
WEEK 3	20m Advanced Nature Study Afternoon Work - Lesson 3 → FIELD WORK Spend an extended time on field work.	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 16 <i>Section 1-3</i> Materials: Master of Field Arts → READ p.47 → READ & PRACTICE Continue to practice your warm-up exercises and then learn about perspective. ∞ Article Link: 12 Simple Warm-Up Exercises ∞ Article Link: 5 Great Exercises to Learn Perspective	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 17 Materials: Journal → JOURNAL Enjoy!	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 18 Materials: Journal → READ & PRACTICE Here are two articles to read and work through to practice your pencil sketching skills. ∞ Article Link: The 15-minutes-a-Day Drawing Exercise Routine ∞ Article Link: How to Draw Rocks → JOURNAL Enjoy!	• PLAN WEEKLY special study observations citizen science observations extended journaling time

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 4	45m Advanced Nature Study Lessons - Lesson 19 Materials: Journal → VIEW Watch another example of using dots for mapping out a sketch. ∞ Article Link: Connect the Dots → JOURNAL Try out the dot mapping technique in the field and see if you find it helpful.	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 20 Materials: Journal → JOURNAL Enjoy!	
WEEK 4	20m Advanced Nature Study Afternoon Work - Lesson 4 → <i>FIELD WORK</i> Spend an extended time on field work.	
WEEK 5	45m Advanced Nature Study Lessons - Lesson 21 <i>Section 1-3</i> Materials: Master of Field Arts, Journal → READ p.48-51 → PRACTICE Make a chart like the one on p.48 to practice your pencil skills. Label the categories on the left-hand side and the graduation from light to dark at the top. Then fill in each square. → JOURNAL Make gesture drawings of many different things today using quick, light pencil lines. → VIEW Watch a few minutes of live gesture drawing. ∞ Video Link: Gesture Sketching in Nature	• EXTRA HELPINGS If you haven't used a blending stump with pencil work before, you might want to experiment with one. Any art store will carry one.
WEEK 5	45m Advanced Nature Study Lessons - Lesson 22 Materials: Journal → JOURNAL Enjoy!	
WEEK 5	45m Advanced Nature Study Lessons - Lesson 23 Materials: Journal → VIEW ∞ Video Link: Four Drawing Tips → JOURNAL Practice gesture drawing and the other drawing tips from John Muir Laws.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 5	45m Advanced Nature Study Lessons - Lesson 24 Materials: Journal → JOURNAL Enjoy!	
WEEK 5	45m Advanced Nature Study Lessons - Lesson 25 Materials: How to Teach Nature Journaling, Journal → READ OR VIEW p.47-49 ∞ Video Link: NJC Zoom In Zoom Out → JOURNAL Focus on zooming in and zooming out in your journal today.	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 5	20m Advanced Nature Study Afternoon Work - Lesson 5 → <i>FIELD WORK</i> Spend an extended time on field work.	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 26 <i>Section 1-3</i> Materials: Master of Field Arts, Journal → INTRO This week, we will learn about ink and pen types. Make some notes as you read about any new types of pens and ink that you want to experiment with in your journal. → READ p.53-58 → VIEW ∞ Video Link: How Printing Ink is Made → JOURNAL Experiment in your nature journal with whatever pens you have available.	• EXTRA HELPINGS If interested, take a look at this review of a classic book on pen and ink drawing. ∞ Video Link: Book Review of Rendering in Pen and Ink
WEEK 6	45m Advanced Nature Study Lessons - Lesson 27 Materials: Journal → VIEW ∞ Video Link: NJC Asking Questions → JOURNAL Focus on asking questions today. Practice the skill of curiosity and see what you find!	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 28 <i>Section 1-3</i> Materials: Master of Field Arts → READ p.147-154 → EXPERIMENT Try making your own pen and ink. If walnut shells, charcoal, or feathers are hard to come by, then here is an article that gives instructions for making ink from raspberries, tea, or coffee and a straw. ∞ Article Link: Create Your Own Ink	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 6	45m Advanced Nature Study Lessons - Lesson 29 Materials: Journal → VIEW ∞ Video Link: NJC Fast Sketching Tips → JOURNAL Try out some of the new things you have learned as you journal today.	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 30 Materials: Journal → JOURNAL Enjoy!	
WEEK 6	20m Advanced Nature Study Afternoon Work - Lesson 6 → <i>FIELD WORK</i> Spend an extended time on field work.	
WEEK 7	45m Advanced Nature Study Lessons - Lesson 31 <i>Section 1-3</i> Materials: Master of Field Arts, Journal, Various Pens → READ p.58-60 → PRACTICE Copy all of the Slow and Deliberate Strokes (p.59) in your journal. Take your time and pay	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

attention to how your pen works. Try out different pens to compare the line-making process and effects. Then work on the Quick and Free Strokes (p.60).

→ JOURNAL

Try out different pens and strokes while sketching.

WEEK 7 45m Advanced Nature Study Lessons - Lesson 32

 Materials: Journal

→ VIEW

∞ Video Link: NJC Nature Comics

→ JOURNAL

Look for nature events or stories to document in your journal.

WEEK 7 45m Advanced Nature Study Lessons - Lesson 33

 Materials: Journal

→ JOURNAL

Enjoy!

• PLAN WEEKLY

-  special study observations
-  citizen science observations
-  extended journaling time

WEEK 7 45m Advanced Nature Study Lessons - Lesson 34

 Materials: How to Teach Nature Journaling, Journal

→ READ OR VIEW

p.54-60

∞ Video Link: NJC Nature Collection

→ JOURNAL

Focus on making a collection of items today.

WEEK 7 45m Advanced Nature Study Lessons - Lesson 35

 Materials: Journal

→ JOURNAL

Enjoy!

WEEK 7 20m Advanced Nature Study Afternoon Work - Lesson 7

→ FIELD WORK

Spend an extended time on field work.

WEEK 8 45m Advanced Nature Study Lessons - Lesson 36

Section 1-3

 Materials: Master of Field Arts

→ READ

p.60-63

→ PRACTICE

Starting on p.60 with the Ink Tone, Texture, and Vegetation Palettes, practice each technique demonstrated. Make the charts and add your own additional textures. Then work on drawing with the outline and broken line techniques. Make several positive and negative space sketches of similar items. It takes some time to get good at seeing the shapes! Then, put the elements together and work on three versions of the same scene as Roseann did on p.62.

WEEK 8 45m Advanced Nature Study Lessons - Lesson 37

 Materials: Journal

→ JOURNAL

Enjoy!

WEEK 8 45m Advanced Nature Study Lessons - Lesson 38

 Materials: How to Teach Nature Journaling, Journal

→ READ

p.180-182

→ JOURNAL

Focus on drawing the insides and outsides of things.

• PLAN WEEKLY

-  special study observations
-  citizen science observations
-  extended journaling time

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 8	45m Advanced Nature Study Lessons - Lesson 39 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS Have you considered using a fountain pen? ∞ Article Link: The Gift of a Fountain Pen
WEEK 8	45m Advanced Nature Study Lessons - Lesson 40 Materials: How to Teach Nature Journaling, Journal → READ p.183-186 → JOURNAL Focus on showing the structure of a thing.	
WEEK 8	20m Advanced Nature Study Afternoon Work - Lesson 8 → <i>FIELD WORK</i> Spend an extended time on field work.	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 41 <i>Section 1-3</i> Materials: Master of Field Arts, Journal → INTRO Today, we will start reading about adding the fun variation of map-making to your journaling toolbox. There are so many ways to be creative with maps! Find ways that you enjoy! → READ p.65-73 → VIEW Explore the Mappa Mundi. Also, take a look at the Magna Carta while you are there! ∞ Website Link: Mappa Mundi → JOURNAL Pick a map type to explore, documenting a past trip or plan a new adventure.	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 42 Materials: How to Teach Nature Journaling, Journal → READ OR VIEW p.107-110 ∞ Video Link: NJC Mapping → JOURNAL Focus on mapping somewhere local to you.	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 43 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 9	45m Advanced Nature Study Lessons - Lesson 44 Materials: Journal → VIEW ∞ Article Link: Drawn walks → JOURNAL Focus on making a map of a walk. Stop and journal along the way.	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 45 Materials: Journal → JOURNAL Enjoy!	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 9	20m Advanced Nature Study Afternoon Work - Lesson 9 → FIELD WORK Spend an extended time on field work.	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 46 Section 1-3 Materials: Master of Field Arts, Journal → READ p.74-75 → JOURNAL Try making a strip map to add to your journal.	• EXTRA HELPINGS View some more medieval maps made by Matthew Paris from the British Library. ∞ Article Link: The Maps of Matthew Paris
WEEK 10	45m Advanced Nature Study Lessons - Lesson 47 Materials: Journal → JOURNAL Enjoy!	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 48 Materials: How to Teach Nature Journaling, Journal → READ p.111-114 → VIEW ∞ Video Link: NJC Cross Section → JOURNAL Find a place to journal about a landscape transition or a cross-section.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 10	45m Advanced Nature Study Lessons - Lesson 49 Materials: Journal → JOURNAL Enjoy!	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 50 Materials: Journal → READ & VIEW ∞ Website Link: Seeing the Unseen → JOURNAL See if you can find an interesting place to use a cut-away in your journal,	
WEEK 10	20m Advanced Nature Study Afternoon Work - Lesson 10 → FIELD WORK Spend an extended time on field work.	
WEEK 11	45m Advanced Nature Study Lessons - Lesson 51 Section 1-3 Materials: Master of Field Arts, Journal → READ p.76-79 → JOURNAL Come up with an interesting way to use behavior and data maps.	
WEEK 11	45m Advanced Nature Study Lessons - Lesson 52 Materials: Journal → VIEW ∞ Video Link: NJC Sampling and Graphing → JOURNAL Focus on trying out some of John Muir Laws' ideas in your journal.	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 11 45m Advanced Nature Study Lessons - Lesson 53 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 11 45m Advanced Nature Study Lessons - Lesson 54 Materials: How to Teach Nature Journaling, Journal → READ p.198-205 → JOURNAL Explore more ways to use numbers, charts, and graphs in your journal.	
WEEK 11 45m Advanced Nature Study Lessons - Lesson 55 Materials: Journal → JOURNAL Enjoy!	
WEEK 11 20m Advanced Nature Study Afternoon Work - Lesson 11 → FIELD WORK Spend an extended time on field work.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 56 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 57 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 58 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 12 45m Advanced Nature Study Lessons - Lesson 59 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 60 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 20m Advanced Nature Study Afternoon Work - Lesson 12 → EXAMS Answer exam questions from: Advanced Nature Study Lessons	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 1

45m Advanced Nature Study Lessons - Lesson 61

Materials: Links, paints, paint brushes, journal

→ INTRO

We will spend the next few weeks exploring the wonderful world of color! If you are new to adding color to your nature journal, Roseann's article will introduce you to the basics. If you are not new, consider trying out Roseann's palette of colors.

→ READ

∞ Article Link: Meet the Minimalist Paint Set

→ PRACTICE

If you haven't made a chart of how your particular colors work when mixing them, make a chart like in the article and experiment with your colors. It takes a while to get good at knowing your pigments and how to make the colors you need to match what you are looking at in nature. As Roseann points out, some paints are harder to use than others due to what is in them. If you are struggling, consider using the ones Roseann recommends.

Follow what Roseann has put together for working on secondary colors, too.

∞ PDF Link: Color worksheets

WEEK 1

45m Advanced Nature Study Lessons - Lesson 62

Sections 4-6

Materials: Master of Field Arts

→ INTRO

People have been making paint from the earth for thousands of years. Notice what colors of dirt and rock are around you. Can you reproduce the colors with your paints? With appropriate precautions, you might want to experiment with making your own paints!

→ READ

p.81-top of p.86

→ VIEW

∞ Video Link: Earth Pigments

∞ Video Link: Lost Kingdoms of the Amazon (51:37)

Start at the 51:37 minute mark in this fascinating documentary on the lost civilizations of the Amazon to see some amazing rock art! The rest of it is worth watching as well at another time.

WEEK 1

45m Advanced Nature Study Lessons - Lesson 63

Materials: Journal

→ JOURNAL

Enjoy!

• PLAN WEEKLY

- special study observations
- citizen science observations
- extended journaling time

WEEK 1

45m Advanced Nature Study Lessons - Lesson 64

Materials: How to Teach Nature Journaling, Journal

→ READ OR VIEW

p.80-83

∞ Video Link: NJC Sound Map

→ JOURNAL

Enjoy making a sound map today!

WEEK 1

45m Advanced Nature Study Lessons - Lesson 65

Materials: Journal

→ JOURNAL

Enjoy!

WEEK 1

20m Advanced Nature Study Afternoon Work - Lesson 13

→ FIELD WORK

Spend an extended time on field work.

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 2

45m Advanced Nature Study Lessons - Lesson 66

Sections 4-6

Materials: Master of Field Arts, Journal

→ READ

p.86-88

→ READ

∞ Article Link: Green Earth Pigments

Read/skim this interesting article on green.

→ JOURNAL

Enjoy using colors in your journal!

WEEK 2

45m Advanced Nature Study Lessons - Lesson 67

Materials: Journal

→ JOURNAL

Enjoy!

WEEK 2

45m Advanced Nature Study Lessons - Lesson 68

Materials: Journal

→ INTRO

Explore more paints and inks to make from nature.

→ VIEW

∞ Article Link: How to Make Plant Paints

∞ Article Link: Flowers as Crayons

∞ Article Link: Making Ink from Berries

∞ Article Link: How to Make Lake Pigments

→ JOURNAL

∞ Podcast Link: Nature Spots

Have fun making some nature spots!

• PLAN WEEKLY

- ☐ special study observations
- ☐ citizen science observations
- ☐ extended journaling time

WEEK 2

45m Advanced Nature Study Lessons - Lesson 69

Materials: How to Teach Nature Journaling, Journal

→ READ OR VIEW

p.76-79

∞ Video Link: NJC Birdsong

→ JOURNAL

Enjoy capturing birdsong in your journal!

WEEK 2

45m Advanced Nature Study Lessons - Lesson 70

Sections 4-6

Materials: Journal

→ JOURNAL

Enjoy!

WEEK 2

20m Advanced Nature Study Afternoon Work - Lesson 14

→ FIELD WORK

Spend an extended time on field work.

WEEK 3

45m Advanced Nature Study Lessons - Lesson 71

Sections 4-6

Materials: Master of Field Arts

→ READ

p.89-95

→ VIEW

Roseann presents a webinar covering what is in the chapter. Skip through, if you wish, until you get to where you can watch her mix her colors and paint the scene

∞ Video Link: Ancient Ochres

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 3	45m Advanced Nature Study Lessons - Lesson 72 Materials: Journal → JOURNAL Enjoy!	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 73 <i>Sections 4-6</i> Materials: Master of Field Arts → VIEW Roseann's webinar will take about two days to watch, so watch 45 minutes of it today. It is really helpful to see how others sketch and paint. See what tips you can pick up and use in your journaling. ∞ Video Link: Quickly Capture Complex Subjects and Add Color	• PLAN WEEKLY special study observations citizen science observations extended journaling time
WEEK 3	45m Advanced Nature Study Lessons - Lesson 74 <i>Sections 4-6</i> Materials: Master of Field Arts → VIEW Finish watching the webinar from yesterday. There will be about a 20-minute section on color mixing that will be similar to what you read earlier in the chapter. Feel free to skip that part. ∞ Video Link: Quickly Capture Complex Subjects and Add Color	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 75 Materials: Journal → JOURNAL Enjoy!	
WEEK 3	20m Advanced Nature Study Afternoon Work - Lesson 15 → FIELD WORK Spend an extended time on field work.	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 76 Materials: Phenology Wheel Template, Journal → INTRO Hopefully, your special studies and citizen science projects are going well. Even if it is fall or winter outside, there are interesting projects to be involved in and creative nature journaling opportunities available. One is a phenology wheel of monthly colors. Building on the nature spots idea, try capturing more of the main colors outside during the month in your journal or on the wheel template, which you can paste into your journal. Then, you can continue the process all year. It might be the color of the sky, snow, and the green of an evergreen. Look around over several days and see what colors you notice. ∞ PDF Link: Phenology Wheel template Look at other examples of interesting things to use a phenology wheel to record. ∞ Website Link: Phenology Wheels	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 77 Materials: Journal → JOURNAL Enjoy!	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 78 Materials: How to Teach Nature Journaling → READ OR VIEW p.39-41 ∞ Video Link: NJC My Secret Plant → JOURNAL You will need a friend to play the secret plant game with today. Have fun!	• PLAN WEEKLY special study observations citizen science observations extended journaling time

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 4 45m Advanced Nature Study Lessons - Lesson 79

Materials: Journal

→ VIEW

∞ Video Link: NJC Little Landscapes

→ JOURNAL

Try out some little landscapes of your own.

WEEK 4 45m Advanced Nature Study Lessons - Lesson 80

Materials: Journal

→ JOURNAL

Enjoy!

WEEK 4 20m Advanced Nature Study Afternoon Work - Lesson 16

→ FIELD WORK

Spend an extended time on field work.

WEEK 5 45m Advanced Nature Study Lessons - Lesson 81

Sections 4-6

Materials: Master of Field Arts, Journal

→ VIEW & READ

∞ Video Link: Weather 101 (Up to the 23:54 minute mark and then 1:12:51-1:39)
Roseann will walk through the reading for today, as well as how to make a cloud chart. p.97-101

→ JOURNAL

Roseann used the Clouds by Altitude chart in the links as her guide for making a cloud chart for her journal. There is a video from another source that also has a version of a cloud chart. Choose a format you like and begin work on your cloud chart. You can use watercolor with wax paper or crayons to draw the clouds as Roseann did first. Or, you can use toned paper and white gouache to paint in the cloud types.

∞ Image Link: Clouds by Altitude
∞ Video Link: The 10 Main Clouds

WEEK 5 45m Advanced Nature Study Lessons - Lesson 82

Materials: How to Teach Nature Journaling, Journal

→ READ OR VIEW

p.45-46

∞ Video Link: NJC Comparison

→ JOURNAL

Focus on things to compare.

WEEK 5 45m Advanced Nature Study Lessons - Lesson 83

Materials: Journal, pencil

→ VIEW

∞ Video Link: Two Quick Techniques for Painting Clouds

→ JOURNAL

Focus on drawing your clouds today.

• PLAN WEEKLY

- special study observations
- citizen science observations
- extended journaling time

WEEK 5 45m Advanced Nature Study Lessons - Lesson 84

Materials: Journal

→ VIEW

The first 45 minutes of this webinar are on the skies. The second half is on drawing water. If interested, you can watch the second half at another time.

∞ Video Link: How to Draw and Paint Water and Sky

→ JOURNAL

Focus on experimenting with more ways to capture your skies in your journal.

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 5	45m Advanced Nature Study Lessons - Lesson 85 Materials: Journal → JOURNAL Enjoy!	
WEEK 5	20m Advanced Nature Study Afternoon Work - Lesson 17 → FIELD WORK Spend an extended time on field work.	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 86 <i>Sections 4-6</i> Materials: Master of Field Arts, Journal → VIEW & READ ∞ Video Link: Weather 101 (23:54-40:00) p.102-middle of p.106 → JOURNAL Focus on noticing the signs around you of what weather is on the way. Jot down some notes. If interested, this article adds more signs of the coming weather based on the cloud type. ∞ Article Link: The 10 Basic Types of Clouds	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 87 <i>Sections 4-6</i> Materials: Master of Field Arts, Journal → READ ∞ Article Link: Cloud Catcher → JOURNAL Today, use your Perspex Palette that was on the supply list for the course to make drawing clouds easier! You don't have to keep looking up and down to draw. You can place your notebook right to the side of the palette and draw the clouds. → VIEW ∞ Video Link: Full Day Time Lapse Watch a full day of clouds in 45 seconds!	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 88 → VIEW ∞ Website Link: Drawing Weather Take some time to observe this week, depending on your weather, what rain, dew, fog, and rainbows actually look like. It is easy to have preconceived ideas of what these things look like, which will make it hard for us to draw what we see.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 6	45m Advanced Nature Study Lessons - Lesson 89 Materials: Journal → VIEW ∞ Video Link: NJC Pace Yourself → JOURNAL Focus on finding your pace and measuring things using your numbers.	
WEEK 6	45m Advanced Nature Study Lessons - Lesson 90 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS For those of you with snow and ice outside at some point in the year, take a look at John Muir Laws demonstrating how to capture that in your journal. ∞ Video Link: Drawing in Winter
WEEK 6	20m Advanced Nature Study Afternoon Work - Lesson 18 → FIELD WORK Spend an extended time on field work.	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 7	45m Advanced Nature Study Lessons - Lesson 91 Sections 4-6 Materials: Master of Field Arts, Journal → VIEW & READ ∞ Video Link: Weather 101 (40:49-47:05) p.106-108 → JOURNAL Be sure to cut out p.174 and secure it in your journal so that you can refer to the weather symbols and the Beaufort wind scale when needed. Then use the tools you have or purchase what you need to add more weather data to your journal. Try estimating the wind direction with the tips on p.108. If you are looking for a Citizen Science Project related to tracking the weather, check out CoCoRaHS, which stands for Community Collaborative Rain, Hail, Snow Network. ∞ Website Link: CoCoRaHS	
WEEK 7	45m Advanced Nature Study Lessons - Lesson 92 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS If interested, here is a longer PDF Roseann has on her site that gets into the variations of clouds. Fascinating! ∞ PDF Link: The Ten Different Types of Clouds
WEEK 7	45m Advanced Nature Study Lessons - Lesson 93 Sections 4-6 Materials: Master of Field Arts, Journal → VIEW Roseann and a few others started a Skyscapito Appreciation Society, which held several Zoom meetings about all things related to the skies. Take some time to watch one or a part of one that interests you. Scroll down to the very bottom first, and then you will see the meetings available in order as you scroll up. ∞ Video Link: Skyscapito Appreciation Society → JOURNAL Focus on something you learned from the Zoom meeting.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 7	45m Advanced Nature Study Lessons - Lesson 94 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS Here is a handy weather report form that you might find useful. ∞ PDF Link: Weather Report
WEEK 7	45m Advanced Nature Study Lessons - Lesson 95 Materials: Journal → VIEW ∞ Video Link: NJC Measure the Wind → JOURNAL Focus on noticing the wind and trying to measure its direction and speed.	
WEEK 7	20m Advanced Nature Study Afternoon Work - Lesson 19 → <i>FIELD WORK</i> Spend an extended time on field work.	
WEEK 8	45m Advanced Nature Study Lessons - Lesson 96 Sections 4-6 Materials: Master of Field Arts, Journal → READ p.109-111 → JOURNAL Create a Cyanometer to keep in your journal to measure the blueness of your sky. Alternate instructions here: ∞ Website Link: Summer Cyanometers	• EXTRA HELPINGS If interested, there is a cyanometer app for iPhones that might be interesting to experiment with. ∞ Website Link: Cyanometer App The cyanometer has been used in a modern-day sculpture. ∞ Website Link: Cyanometer in Slovenia

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 8	45m Advanced Nature Study Lessons - Lesson 97 Materials: Journal → JOURNAL Enjoy!	
WEEK 8	45m Advanced Nature Study Lessons - Lesson 98 Materials: Journal → VIEW ∞ Video Link: NJC Making Mistakes → JOURNAL Don't allow a fear of making a mistake to keep you from getting all your observations down in your journal. Making mistakes and learning from them are part of the process of discovery!	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 8	45m Advanced Nature Study Lessons - Lesson 99 Materials: Journal → INTRO When the weather is not conducive to being outside, you can still nature journal virtually. Roseann has many interesting virtual field trips available on her website. Today we are going to go to Mars! It will take two days to work through the field trip. Stop the video and make your journal entries as Roseann is making hers. → VIEW & JOURNAL ∞ Website Link: Let's Explore Mars!	• EXTRA HELPINGS Something that you may be able to draw looking out your window is the moon! Roseann has a wonderful moon phase chart from a local hand-drawn book that you could use as a basis for drawing your own. Observe each night and draw what you see. Also, there is some great moon information at the NASA site. ∞ Image Link: Moon Phase Chart (scroll down) ∞ Website Link: Moon Phases
WEEK 8	45m Advanced Nature Study Lessons - Lesson 100 Materials: Journal → VIEW & JOURNAL ∞ Website Link: Let's Explore Mars! Finish your field trip from yesterday.	
WEEK 8	20m Advanced Nature Study Afternoon Work - Lesson 20 → FIELD WORK Spend an extended time on field work.	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 101 <i>Sections 4-6</i> Materials: Master of Field Arts → VIEW & REVIEW ∞ Video Link: Animal Tracking 101 Watch the first 46 minutes. We will watch more next week. p.113-118	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 102 Materials: Journal → JOURNAL Enjoy! See if you can find some tracks today!	• EXTRA HELPINGS If last week's Mars field trip got you interested in journaling things in the sky, here is more from Roseann about that topic. ∞ Website Link: Journaling Night Skies
WEEK 9	45m Advanced Nature Study Lessons - Lesson 103 Materials: How to Teach Nature Journaling, Journal → READ OR VIEW p.69-71 ∞ Video Link: NJC Animals on the Move → JOURNAL Find some moving creatures to draw in your journal.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 9	45m Advanced Nature Study Lessons - Lesson 104 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS There is a great kids' book by Jim Arnosky that your library might have called <i>Wild Tracks!</i> that shows lots of animal tracks, life-size. It is very helpful for identification. ∞ Website link: Wild Tracks!
WEEK 9	45m Advanced Nature Study Lessons - Lesson 105 → VIEW ∞ Video Link: Drawing a Badger Skull	
WEEK 9	20m Advanced Nature Study Afternoon Work - Lesson 21 → FIELD WORK Spend an extended time on field work.	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 106 <i>Sections 4-6</i> Materials: Master of Field Arts, Journal → VIEW & REVIEW ∞ Video Link: Animal Tracking 101 Watch from 46 minutes to 1:02:41. We will finish it next week. p.119-middle of p.125 → JOURNAL See if you can read the story of the signs of animals that you find.	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 107 Materials: Journal → JOURNAL Enjoy!	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 108 Materials: How to Teach Nature Journaling, Journal → READ OR VIEW p.97-102 ∞ Video Link: NJC Investigation → JOURNAL Have fun doing some investigation outside and record your observations, questions, and discoveries.	• PLAN WEEKLY special study observations citizen science observations extended journaling time
WEEK 10	45m Advanced Nature Study Lessons - Lesson 109 Materials: Journal → JOURNAL Enjoy!	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 110 Materials: Journal → VIEW ∞ Video Link: The Colours of the Planets → JOURNAL Spend some time journaling the color of the planets! Look up more information about what makes the colors if interested.	
WEEK 10	20m Advanced Nature Study Afternoon Work - Lesson 22 → FIELD WORK Spend an extended time on field work.	

Science: Advanced Nature Study

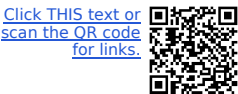
[Click THIS text or scan the QR code for links.](#)



Term 2

WEEK 11	45m Advanced Nature Study Lessons - Lesson 111 <i>Sections 4-6</i>	
	Materials: Master of Field Arts	
	→ VIEW & READ ∞ Video Link: Animal Tracking 101 Watch from 1:02:41 to the end. You might have to finish at another time, but there are some helpful questions and comments shared near the end. p.125-131, 25	
WEEK 11	45m Advanced Nature Study Lessons - Lesson 112	
	Materials: Journal, Pens/Pencil, Watercolors, Watercolor Brush	
	→ JOURNAL There are lots of things to test out from yesterday's webinar. Use your Perspex palette for drawing tracks. Then, practice using pen and ink as well as watercolor to finish the drawing. Next, try using wax paper to record a track trail. Look back at p.25 in your book to see how to mix up the plaster of Paris to make a mold of any tracks you find.	
WEEK 11	45m Advanced Nature Study Lessons - Lesson 113 <i>Sections 4-6</i>	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
	Materials: Master of Field Arts, Journal	
	→ JOURNAL Spend time looking for and investigating scat today!	
WEEK 11	45m Advanced Nature Study Lessons - Lesson 114	
	Materials: How to Teach Nature Journaling, Journal	
	→ READ & VIEW p.198-203 ∞ Video Link: NJC Estimating Numbers	
	→ JOURNAL See what you can find to estimate and use numbers to describe.	
WEEK 11	45m Advanced Nature Study Lessons - Lesson 115	
	Materials: Journal	
	→ JOURNAL Enjoy!	
WEEK 11	20m Advanced Nature Study Afternoon Work - Lesson 23 → FIELD WORK Spend an extended time on field work.	
WEEK 12	45m Advanced Nature Study Lessons - Lesson 116 <i>Exam Week</i>	
	→ EXAMS Answer question(s) related to the course.	
WEEK 12	45m Advanced Nature Study Lessons - Lesson 117 <i>Exam Week</i>	
	→ EXAMS Answer question(s) related to the course.	
WEEK 12	45m Advanced Nature Study Lessons - Lesson 118 <i>Exam Week</i>	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
	→ EXAMS Answer question(s) related to the course.	
WEEK 12	45m Advanced Nature Study Lessons - Lesson 119 <i>Exam Week</i>	
	→ EXAMS	

Science: Advanced Nature Study



[Click THIS text or scan the QR code for links.](#)

Term 2	
Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 120 Exam Week → EXAMS Answer question(s) related to course.	
WEEK 12 20m Advanced Nature Study Afternoon Work - Lesson 24 → EXAMS Answer exam questions from: Advanced Nature Study Lessons	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 1	45m Advanced Nature Study Lessons - Lesson 121 <i>Section 7</i> Materials: Master of Field Arts, Journal → READ p.133-141 → JOURNAL Focus on using words to describe what you are observing.	• EXTRA HELPINGS For more natural history reading suggestions, check out the Alveary Bookshelf for HS Science and Natural History titles. ∞ Website Link: Alveary Bookshelf
WEEK 1	45m Advanced Nature Study Lessons - Lesson 122 Materials: Journal → JOURNAL Enjoy!	
WEEK 1	45m Advanced Nature Study Lessons - Lesson 123 Materials: How to Teach Nature Journaling, Journal → READ OR VIEW p.138-142 ∞ Video Link: NJC Using Words → JOURNAL Explore how adding words can be helpful to your thinking process.	• PLAN WEEKLY special study observations citizen science observations extended journaling time
WEEK 1	45m Advanced Nature Study Lessons - Lesson 124 Materials: Journal → JOURNAL Enjoy!	
WEEK 1	45m Advanced Nature Study Lessons - Lesson 125 Materials: Journal → VIEW ∞ Video Link: NJC Counting and Measuring → JOURNAL Enjoy looking for things to count and measure.	• EXTRA HELPINGS Here, John Muir Laws shows you how to use a compass clinometer. ∞ Video Link: Using a Compass Clinometer
WEEK 1	20m Advanced Nature Study Afternoon Work - Lesson 25 → FIELD WORK Spend an extended time on field work.	
WEEK 2	45m Advanced Nature Study Lessons - Lesson 126 <i>Section 7</i> Materials: Master of Field Arts, Journal → READ p.142-145 → JOURNAL Focus on your writing today.	
WEEK 2	45m Advanced Nature Study Lessons - Lesson 127 Materials: Journal → JOURNAL Enjoy!	
WEEK 2	45m Advanced Nature Study Lessons - Lesson 128 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY special study observations citizen science observations extended journaling time

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 2	45m Advanced Nature Study Lessons - Lesson 129 Materials: How to Teach Nature Journaling, Journal → READ OR VIEW p.151-153 ∞ Video Link: NJC Writing Poems → JOURNAL Enjoy writing some poems today!	• EXTRA HELPINGS A nice collection of nature poems is helpful if you want to add some to your journal. Your local library may have some nice options as well. ∞ Website Link: The Four Seasons
WEEK 2	45m Advanced Nature Study Lessons - Lesson 130 Materials: Journal → INTRO Emily Dickinson wrote a poem called Bloom. She also put together a huge collection of pressed flowers she collected on her nature walks. Read the poem and view the herbarium that, surprisingly, is still preserved. Then watch the modern creative animation of both put together, and the backstory to its creation. → READ ∞ Poem Link: Bloom ∞ Website Link: Harvard (Emily's Herbarium) ∞ Website Link: Musical Animation of Bloom → JOURNAL Find and add some favorite nature poems to your journal. Create some new ones!	
WEEK 2	20m Advanced Nature Study Afternoon Work - Lesson 26 → FIELD WORK Spend an extended time on field work.	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 131 → VIEW ∞ Website Link: Landscapitos Watch the first 45 minutes.	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 132 → VIEW & PRACTICE ∞ Website Link: Landscapitos Watch the second half and work along with Roseann, stopping the video when needed.	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 133 Materials: Journal → JOURNAL Practice using your perspex palette to make a grid drawing for a landscapito.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 3	45m Advanced Nature Study Lessons - Lesson 134 Materials: Journal → JOURNAL Practice some landscapitos!	
WEEK 3	45m Advanced Nature Study Lessons - Lesson 135 Materials: Journal → VIEW ∞ Video Link: NJC Using Inference to Explore → JOURNAL Enjoy making inferences and looking for evidence!	
WEEK 3	20m Advanced Nature Study Afternoon Work - Lesson 27 → FIELD WORK Spend an extended time on field work.	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 4	45m Advanced Nature Study Lessons - Lesson 136 → VIEW ∞ Website Link: Using Grids: 3D Cube Landscapes Watch the first 45 minutes. Roseann covers the grid method for landscapes again before getting into the 3D cube drawings. She goes into a little more detail than before, so it may be a helpful review.	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 137 Materials: Paper, pencil → VIEW & PRACTICE ∞ Website Link: Using Grids: 3D Cube Landscapes Watch the next 45 minutes and draw along with the group. Stop the video when needed.	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 138 → VIEW & PRACTICE ∞ Website Link: Using Grids: 3D Cube Landscapes Finish the webinar. Think of somewhere local to you that you might want to do a 3D cube drawing.	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 4	45m Advanced Nature Study Lessons - Lesson 139 Materials: Journal → JOURNAL Enjoy!	
WEEK 4	45m Advanced Nature Study Lessons - Lesson 140 Materials: Journal → INTRO Here is more on block diagrams. They are a little tricky to master. Like anything else, if you keep practicing them, they will get easier! → VIEW ∞ Video Link: NJC Block Diagram → JOURNAL Experiment with these types of diagrams.	
WEEK 4	20m Advanced Nature Study Afternoon Work - Lesson 28 → FIELD WORK Spend an extended time on field work.	
WEEK 5	45m Advanced Nature Study Lessons - Lesson 141 → INTRO Today, Roseann will instruct us in the art and science behind various layout strategies for your journal pages. → VIEW & PRACTICE ∞ Video Link: Layout Strategies Watch the first 45 minutes.	
WEEK 5	45m Advanced Nature Study Lessons - Lesson 142 → VIEW & PRACTICE ∞ Video Link: Layout Strategies Watch the second half.	
WEEK 5	45m Advanced Nature Study Lessons - Lesson 143 Materials: Journal → JOURNAL Try out some new ideas for page layouts and see what you think!	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 5	45m Advanced Nature Study Lessons - Lesson 144 Materials: Journal → JOURNAL Enjoy!	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 5 45m Advanced Nature Study Lessons - Lesson 145

Materials: Journal

→ INTRO

How is your year-long study going? I am sure you have collected a lot of great observations. If you still need to do a short-term project, consider what might be growing outside. Try to find something to observe from start to finish, like a caterpillar egg to a butterfly or a seed to a plant making new seeds. Sometimes you can see many stages of a plant at the same time, as John Muir Laws demonstrates in today's video.

→ VIEW

∞ Video Link: NJC Plant Timeline

→ JOURNAL

See what you can find that would work well for a timeline.

WEEK 5 20m Advanced Nature Study Afternoon Work - Lesson 29

→ FIELD WORK

Spend an extended time on field work.

WEEK 6 45m Advanced Nature Study Lessons - Lesson 146

Materials: Journal

→ INTRO

Think about what is right around you that you would like to know more about, and how to draw it in your journal. Look through John Muir Laws' list of topics and find something relevant to you that you are interested in.

→ VIEW & PRACTICE

∞ Video Link: John Muir Laws Blog Archives

→ JOURNAL

Enjoy!

WEEK 6 45m Advanced Nature Study Lessons - Lesson 147

→ VIEW & PRACTICE

∞ Video Link: John Muir Laws Blog Archives

Finish whatever workshop you started yesterday. If it is longer than two days' worth, take another day this week to finish.

WEEK 6 45m Advanced Nature Study Lessons - Lesson 148

Materials: Journal

→ JOURNAL

Enjoy!

• PLAN WEEKLY

- ☐ special study observations
- ☐ citizen science observations
- ☐ extended journaling time

WEEK 6 45m Advanced Nature Study Lessons - Lesson 149

Materials: Journal

→ VIEW

∞ Video Link: NJC Research and Infographics

→ JOURNAL

Find a question you have asked in your journal and research to find some answers. Then make an infographic of what you have found out.

WEEK 6 45m Advanced Nature Study Lessons - Lesson 150

Materials: Journal

→ JOURNAL

Enjoy!

WEEK 6 20m Advanced Nature Study Afternoon Work - Lesson 30

→ FIELD WORK

Spend an extended time on field work.

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 7	45m Advanced Nature Study Lessons - Lesson 151 Materials: Journal, paint, paint brushes, paper, pencil → INTRO Choose a virtual field trip to go on with Roseann and draw and paint along with the group! → VIEW & PRACTICE ∞ Video Link: Roseann's Field Trips	
WEEK 7	45m Advanced Nature Study Lessons - Lesson 152 Materials: Journal, paints, paint brushes, pencil → VIEW & PRACTICE ∞ Video Link: Roseann's Field Trips Finish the field trip you started yesterday. Take your time and stop the video to finish your journaling.	
WEEK 7	45m Advanced Nature Study Lessons - Lesson 153 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 7	45m Advanced Nature Study Lessons - Lesson 154 Materials: Journal → VIEW ∞ Video Link: NJC Stewardship → JOURNAL Find a friend to take with you and do some cleanup somewhere before journaling together.	
WEEK 7	45m Advanced Nature Study Lessons - Lesson 155 Materials: Journal → JOURNAL Enjoy!	
WEEK 7	20m Advanced Nature Study Afternoon Work - Lesson 31 → FIELD WORK Spend an extended time on field work.	
WEEK 8	45m Advanced Nature Study Lessons - Lesson 156 → VIEW & PRACTICE ∞ Video Link: Bumps and Divots	
WEEK 8	45m Advanced Nature Study Lessons - Lesson 157 → VIEW & PRACTICE ∞ Video Link: Bumps and Divots Finish up from yesterday. The group sharing time is always very interesting if you want to keep watching for a third day.	
WEEK 8	45m Advanced Nature Study Lessons - Lesson 158 Materials: Journal → VIEW ∞ Video Link: NJC Biodiversity → JOURNAL Explore the variation around you!	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 8	45m Advanced Nature Study Lessons - Lesson 159 Materials: Journal → JOURNAL Enjoy!	

Science: Advanced Nature Study

Click [THIS text](#) or
scan the QR code
for links.



Term 3

WEEK 8	45m Advanced Nature Study Lessons - Lesson 160 Materials: Journal → JOURNAL Enjoy!	
WEEK 8	20m Advanced Nature Study Afternoon Work - Lesson 32 → FIELD WORK Spend an extended time on field work.	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 161 Materials: Journal → INTRO This week, we will explore the International Nature Journaling Week's 2023 program on the senses. Pick which of the senses you want to explore and click on that day's adventures. Watch the short introductory video at the top and then scroll down to the Nature Journaling Prompts and Ideas. Enjoy! → VIEW ∞ Website Link: International Nature Journaling Week 2023 → JOURNAL Enjoy focusing on one of your senses today!	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 162 → INTRO Pick a second sense to explore today! → VIEW & JOURNAL ∞ Website Link: International Nature Journaling Week 2023	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 163 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY ☐ special study observations ☐ citizen science observations ☐ extended journaling time
WEEK 9	45m Advanced Nature Study Lessons - Lesson 164 Materials: Journal → VIEW ∞ Video Link: NJC Sense of Place → JOURNAL Enjoy hunting for treasures!	
WEEK 9	45m Advanced Nature Study Lessons - Lesson 165 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS If you enjoyed the work of the International Nature Journaling Week, there are other years' adventures to explore, too! ∞ Website Link: International Nature Journaling Week 2020
WEEK 9	20m Advanced Nature Study Afternoon Work - Lesson 33 → FIELD WORK Spend an extended time on field work.	
WEEK 10	45m Advanced Nature Study Lessons - Lesson 166 Materials: Journal → INTRO This week, we will explore some apps that help with identification. See what might be helpful for you. → VIEW ∞ Website Link: Pl@ntNet (Plants and Trees) ∞ Website Link: Merlin Bird ID (Birds) ∞ Website Link: Star Walk 2 ∞ Website Link: SkyView Lite	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

	∞ Website Link: MyRadar (Metadata) → JOURNAL Explore using an app if needed.	
WEEK 10	📅 45m Advanced Nature Study Lessons - Lesson 167 📁 Materials: Journal → INTRO Here are some more apps. See what might be helpful for you. → VIEW ∞ Website Link: All Trails ∞ Website Link: LeafSnap ∞ Website Link: NPS (National Park Service) ∞ Website Link: Day One (Digital Journaling) ∞ Website Link: Nature Journaling with Procreate → JOURNAL Enjoy!	
WEEK 10	📅 45m Advanced Nature Study Lessons - Lesson 168 📁 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY 📅 special study observations 📅 citizen science observations 📅 extended journaling time
WEEK 10	📅 45m Advanced Nature Study Lessons - Lesson 169 📁 Materials: Journal → JOURNAL Enjoy!	• EXTRA HELPINGS If interested, watch a way to animate your nature journaling in Procreate. ∞ Video Link: Animated Nature Journal Page
WEEK 10	📅 45m Advanced Nature Study Lessons - Lesson 170 📁 Materials: Journal → VIEW ∞ Video Link: NJC Drawing Diagrams → JOURNAL Focus on making some diagrams today.	
WEEK 10	📅 20m Advanced Nature Study Afternoon Work - Lesson 34 → FIELD WORK Spend an extended time on field work.	
WEEK 11	📅 45m Advanced Nature Study Lessons - Lesson 171 📁 Materials: Journal → INTRO For our last week of class, choose one of Roseann's Around the World workshops focusing on trees or animals. → VIEW & JOURNAL ∞ Website Link: Nature Journal Academy Around the World	
WEEK 11	📅 45m Advanced Nature Study Lessons - Lesson 172 📁 Materials: Journal → INTRO Continue your Around the World workshop. → VIEW & JOURNAL ∞ Website Link: Nature Journal Academy Around the World	

Science: Advanced Nature Study

[Click THIS text or scan the QR code for links.](#)



Term 3

WEEK 11 45m Advanced Nature Study Lessons - Lesson 173 Materials: Journal → JOURNAL Enjoy!	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 11 45m Advanced Nature Study Lessons - Lesson 174 Materials: Journal → VIEW ∞ Video Link: NJC Metacognition → JOURNAL Enjoy looking back through your work and seeing all that you have done this year! I hope you have grown in your love for journaling, all of creation, and the one true God who has made it all!	
WEEK 11 45m Advanced Nature Study Lessons - Lesson 175 Materials: Journal → JOURNAL Enjoy!	
WEEK 11 20m Advanced Nature Study Afternoon Work - Lesson 35 → FIELD WORK Spend an extended time on field work.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 176 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 177 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 178 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	• PLAN WEEKLY - special study observations - citizen science observations - extended journaling time
WEEK 12 45m Advanced Nature Study Lessons - Lesson 179 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Advanced Nature Study Lessons - Lesson 180 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 20m Advanced Nature Study Afternoon Work - Lesson 36 → EXAMS Answer exam questions from: Advanced Nature Study Lessons	

Science: Advanced Nature Study

Examination

Term 1

Advanced Nature Study Lessons

- Take pictures of 10 of your favorite journaling pages from the term.
- Describe some of the new skills or tools you have been experimenting with this term, and what you think about them. What have you liked and not liked, and where do you want to grow?
- What was one of your most interesting findings or observations from the term? What are you wondering about?

Term 2

Advanced Nature Study Lessons

- Take pictures of 10 of your favorite journaling pages from the term.
- Describe some of the new skills or tools you have been experimenting with this term and what you think about them. What have you liked and not liked, and where do you want to grow?
- What was one of your most interesting findings or observations from the term? What are you wondering about?

Term 3

Advanced Nature Study Lessons

- Take pictures of 10 of your favorite journaling pages from the term.
- Describe some of the new skills or tools you have been experimenting with this term and what you think about them. What have you liked and not liked, and where do you want to grow?
- Describe your year-long study. What plants, animals, etc., did you discover? Describe some of the changes you saw. What were some of the highlights? What surprised you? What questions did it generate? Include a few pictures of your journal pages about your study. Any ideas for other longer-term studies you could do?